

Oleksiy Nedobiychuk

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 Pisa, Italy

Education

- **M.Sc. in Computer Science (AI curriculum), University of Pisa** [2023 – Ongo.] 109/110(e).
- **B.Sc. in Computer Science, University of Pisa** [2019 – 2023] 106/110.
Thesis title: *Realizzazione di una Web Application per la gestione del patrimonio culturale della Regione Toscana.*

Relevant Experience

Software Engineer Intern at TD Group S.R.L, Vecchiano, Italy [Jul. 2025 – Ongo.]:

- Updated and improved agrarial estimation algorithms in **Python**. Enhanced robustness and performance by using caching (20% speedup). Created and integrated web scraping scripts using **BeautifulSoup** and **Playwright** to automate data collection
- Implemented data visualizations using **Highcharts** to present complex agrarian data insights effectively.

Full-Stack Developer at TAI Solutions S.R.L Pisa, Italy [Sept – Nov. 2022]:

- Built a **Spring**-based microservice for tenant management, integrating **Kyecloak** to enable secure authentication and authorization across multiple clients.
- Created a **Angular** web application to manage tenants and users, improving administration efficiency.

Projects

Professional Cycling Analysis - Data Mining Project

- Handled data understanding: assessed data quality, studied data distribution, and analyzed correlations between features. Improved the dataset through data imputation, cleaning, feature engineering, and outlier detection. Identified common characteristics of riders by clustering the data using K-means and DBSCAN algorithms. Built and trained a neural network (NN) model to predict riders' performance, achieving 90% accuracy on the test set.

Neural Network For Regression/Classification Tasks - Machine Learning Project

- Designed and built a flexible neural network (NN) from scratch using Python and NumPy, implementing both forward and backward propagation logic. Introduced Nesterov momentum into the optimizer, as well as L1 and L2 regularization techniques to mitigate overfitting. Developed a grid search algorithm to tune the model's hyperparameters. Achieved a mean squared error of 0.01 on the regression task and an accuracy of 95

ProbLog Based AI Agent

- Developed an AI agent using ProbLog for a 2D 1vs1 fighting game. Assembled a knowledge base (KB) to represent the domain under uncertainty. Enabled the agent to adapt to opponent tactics through dynamic probability estimation. In simulations, the agent outperformed the Monte Carlo Tree Search (MCTS) agent, MinMax agent, and Prolog-based agent.

Web Application - Full-Stack Development

- Developed a fully functional website using Svelte, WebGL2 and vanilla JavaScript for the front end. Designed and integrated a PostgreSQL database into a Django-based back end. Deployed the application on a Debian-based OS using Nginx and Apache web servers.

Skills

Languages	• Fluent in English, Italian, Ukrainian.
Programming	• Proficient in: Python, TypeScript (JS). Familiar with: C++, C. Eager to learn: Rust and Go.
Tools	• PostgreSQL, MongoDB, Django, Angular, Svelte, GIT, Docker.
ML/DS	• PyTorch, TensorFlow, Scikit-learn,Pandas, Numpy, Matplotlib, Seaborn.
Technologies	• Linux (Ubuntu, Arch), Shell scripting, vim, Copilot, ChatGPT.
Soft	• Problem Solving, Adaptability, Teamwork, Deep Focus, Goal Oriented, Critical Thinking.
Interests	• Gym, Reading, Hiking, Roll Skating, Reinforcment Learning.